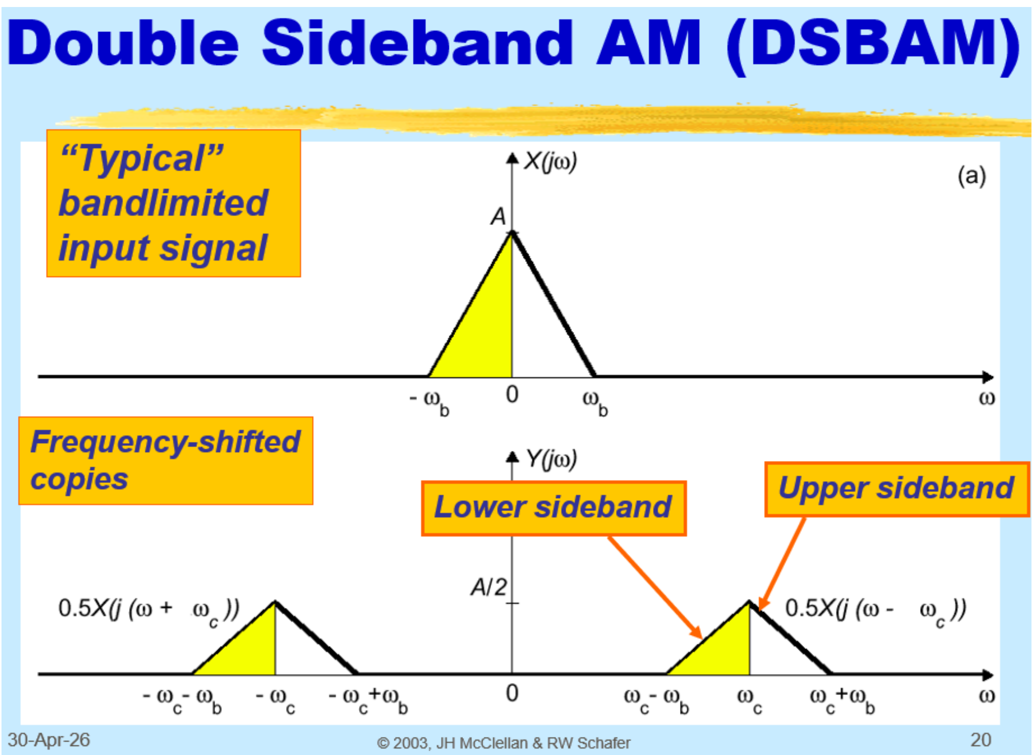
SPRING 2026 - CMPE362 - HOMEWORK 5
Double Sideband Amplitude Modulation (DSBAM)

The aim of this homework is to apply our knowledge on amplitude modulation to a practical example. You are given 6 seconds of a hypothetical radio signal recording `modulated.wav` of three radio channels and you are asked to demodulate and extract audio from each channel.

What we know:

1. The technique used to create this signal is double sideband amplitude modulation (see lecture slide below).
2. There are three radio channels in this broadcast.
3. Before modulation, inputs were bandlimited to [-3kHz, +3kHz].

Recall that for a single input, DSBAM is achieved via y(t) = x(t) * cos(2*pi*f_c*t)



What you need to do is to find the carrier frequencies of each channel and extract the audio located at that channel. If you succeed, you will extract 3 different songs from the three channels. Inspecting the `modulated.wav` via the methods we used throughout our homeworks as well as trial & error should suffice for reaching this goal.

You may hear a very faint bleed out from other channels as the low pass filter used during modulation is not ideal

Submission

Upload:
- All MATLAB scripts (.m files)
- Three extracted audio files (.wav files)
- A presentation (.pdf)

Presentation Format

- (1 slide) Title Page (your name, student ID, etc.)
- (1 slide) Waveforms of each extracted radio channel audio file. (Three in total)
- (1 - 2 slide) What are the carrier frequencies and relevant information about the three channels?

Explain your approach on finding the carrier frequencies of the three radio channels. Add images/figures if you need to.

LLM Disclaimer

You are allowed to use LLMs while doing this homework. However:

Please do not copy large blocks of generated explanations into your slides.
Try to interpret what you observe and write short, clear comments in your own words.

The grading will be done charitably, clarity and engagement matter more than perfection.